PHASE THREE - PROJECT METHODOLOGY

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Industry: Healthcare
Project Title: Improving Maternal Health Insights Through Data Analysis
Proposed Approach: Python-Based Analysis of Maternal Health Data

The project will follow seven practical stages, moving from finding suitable data to producing evidence-based insights and recommendations. The team will use Python to support data preparation, analysis, and visualization.

1. Data Sourcing

Identify a suitable dataset from a credible source, such as a government health-data portal, the World Health Organization, a research repository, or a published study. If suitable real data are unavailable, the team will use a clearly labelled synthetic dataset.

2. Data Collection and Scope Definition

Download or obtain permission to use the dataset, document its source and definitions, and select relevant variables, such as antenatal visits, delivery outcomes, reported complications and maternal outcomes. Use only de-identified data and respect access conditions and privacy requirements.

3. Data Inspection and Understanding

Load the data into Python and examine its structure, column names, data types, number of records and summary statistics. Read the data dictionary, where available, to understand what each variable means and identify limitations before analysis.

4. Data Cleaning and Preparation

Use Pandas and NumPy to check for missing values, duplicate records, inconsistent formats, invalid entries, and possible outliers. Correct issues where justified, document decisions, and avoid inventing values or removing records without a clear reason.

5.Exploratory Data Analysis

Calculate descriptive statistics and examine patterns in the available maternal-health indicators. Compare groups or time periods only where the dataset supports a fair comparison, and identify data-quality gaps that may affect interpretation.

6. Data Visualization and Interpretation

Use Matplotlib to create clear charts and graphs showing important trends, distributions, and missing-data patterns. Interpret the results carefully, distinguishing observed associations from causes and acknowledging limitations in coverage or reporting.

7. Reporting and Recommendations

Present the findings in a concise report or presentation, supported by a reproducible Jupyter Notebook. Recommend practical ways to improve maternal-health data recording, completeness, consistency and use for planning. State what further data or investigation is needed.

Expected output: a documented dataset, a reproducible Python notebook, visualizations, and actionable recommendations. This project is an exploratory data-analysis prototype, not a clinical decision-making tool.